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(54) **MANUFACTURING APPARATUS AND METHOD FOR ABSORBENT BODY**

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13/537 (2013.01); **A61F 2013/15821** (2013.01)

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2013/15821

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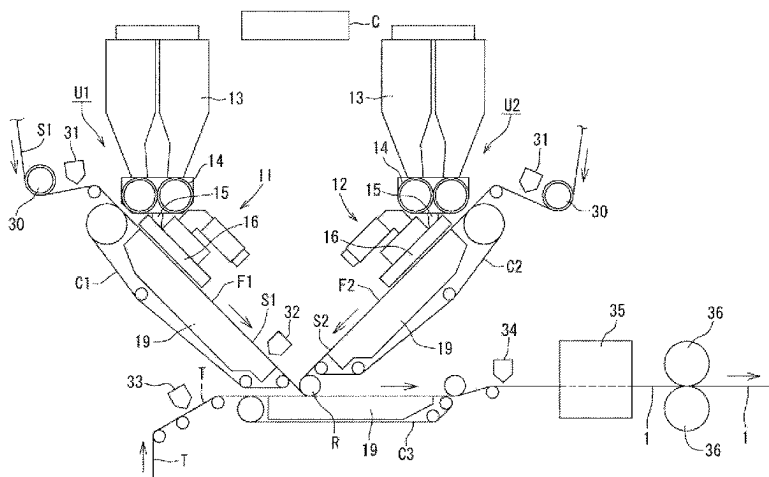
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ABSTRACT

A manufacturing apparatus for an absorbent body including: a first conveyor that conveys an air-permeable first sheet while sucking the first sheet along a flat first slope surface; a second conveyor that conveys an air-permeable second sheet while sucking the second sheet along a flat second slope surface having a slope opposite to the first slope surface; a first distribution device that forms a first absorbent layer in the first sheet by distributing absorbent powder on an upper surface of the first sheet being conveyed along the first slope surface; a second distribution device that forms a second absorbent layer in the second sheet by distributing absorbent powder on an upper surface of the second sheet being conveyed along the second slope surface; and a stacking part that stacks the first sheet conveyed by the first conveyor and the second sheet conveyed by the second conveyor in such a manner as to cause the first absorbent layer of the first sheet and the second absorbent layer of the second sheet to face each other.

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5 Claims, 6 Drawing Sheets



(58) **Field of Classification Search**

USPC 425/81.1

See application file for complete search history.

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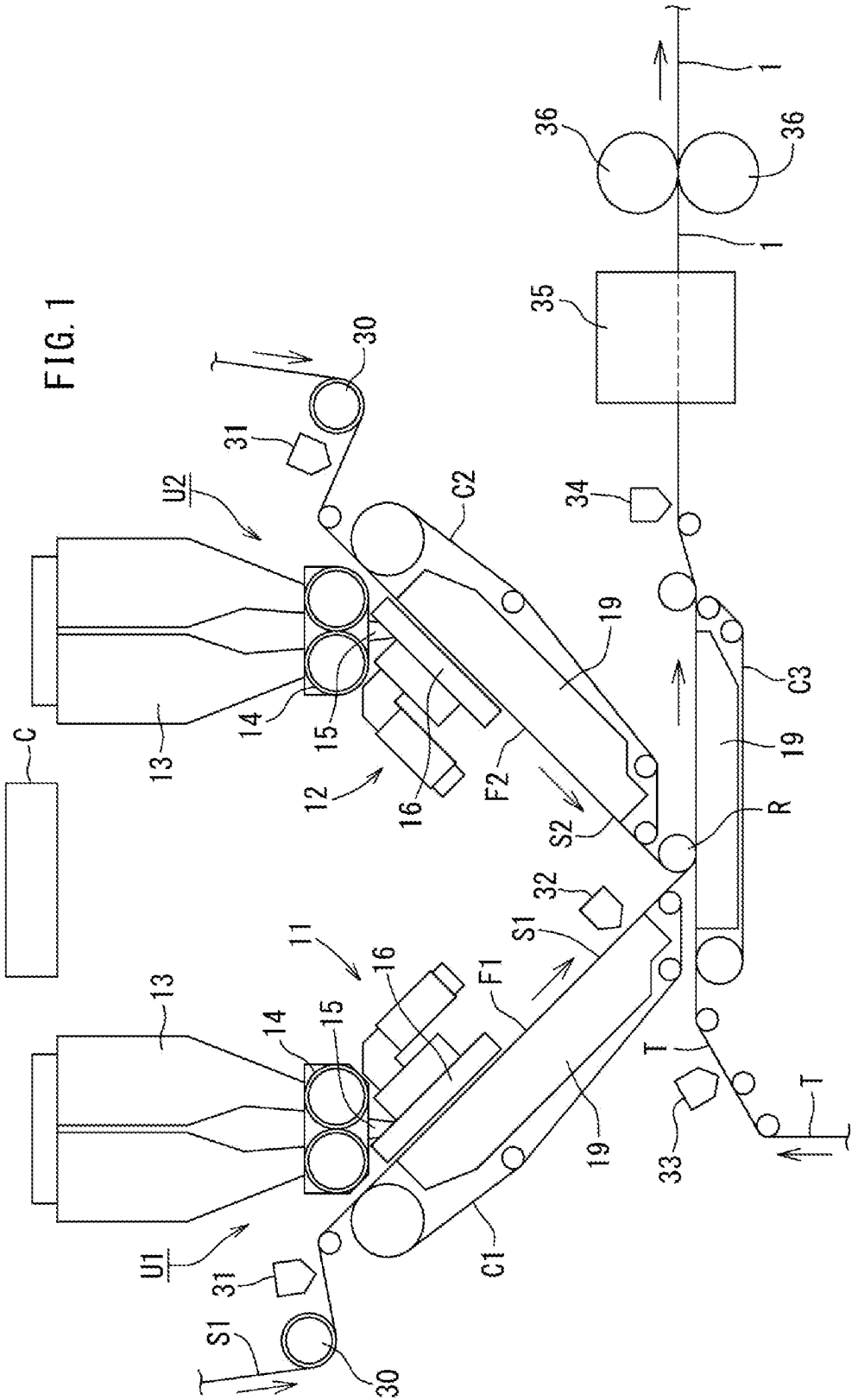


FIG. 2

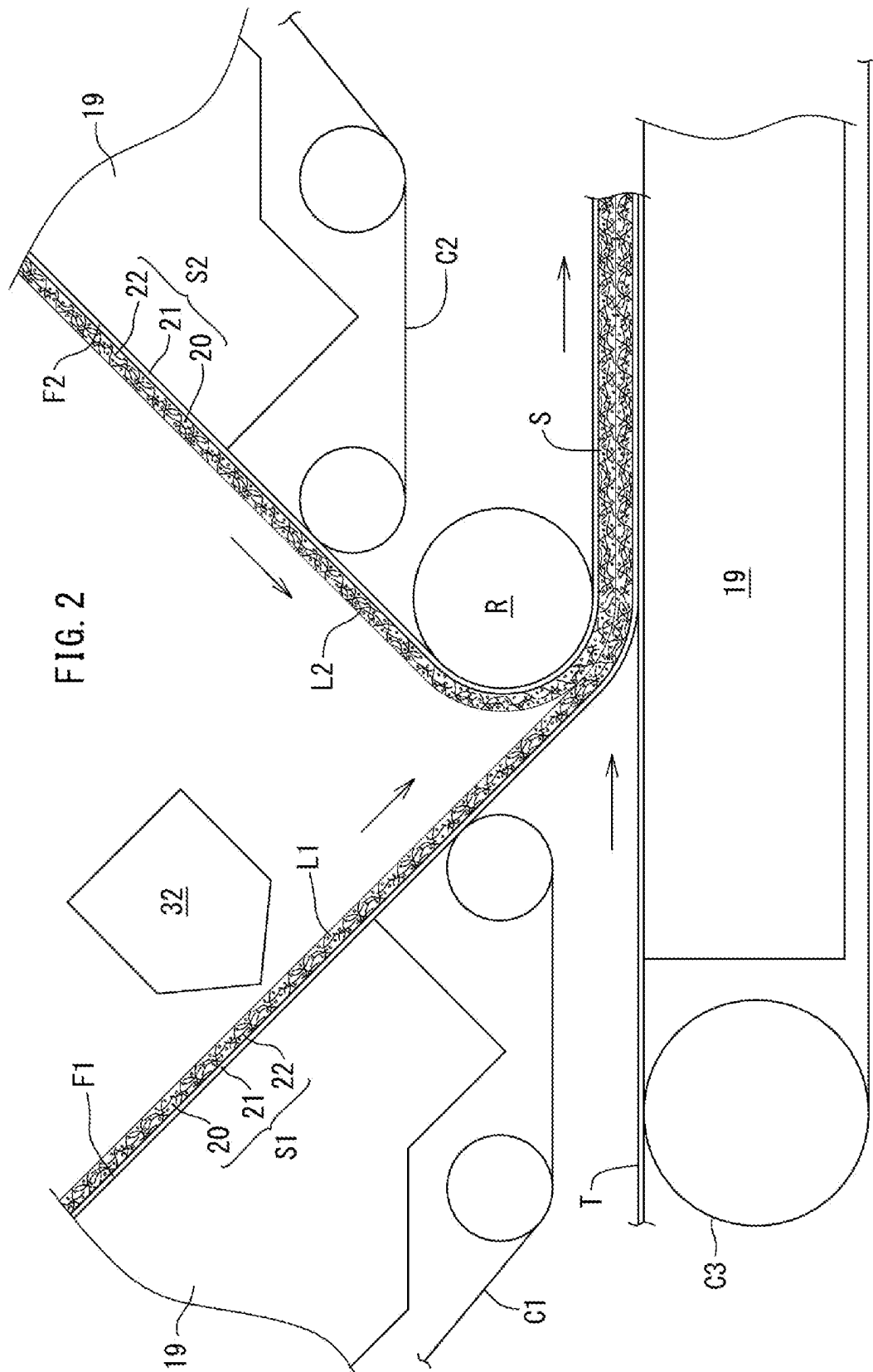


FIG. 3

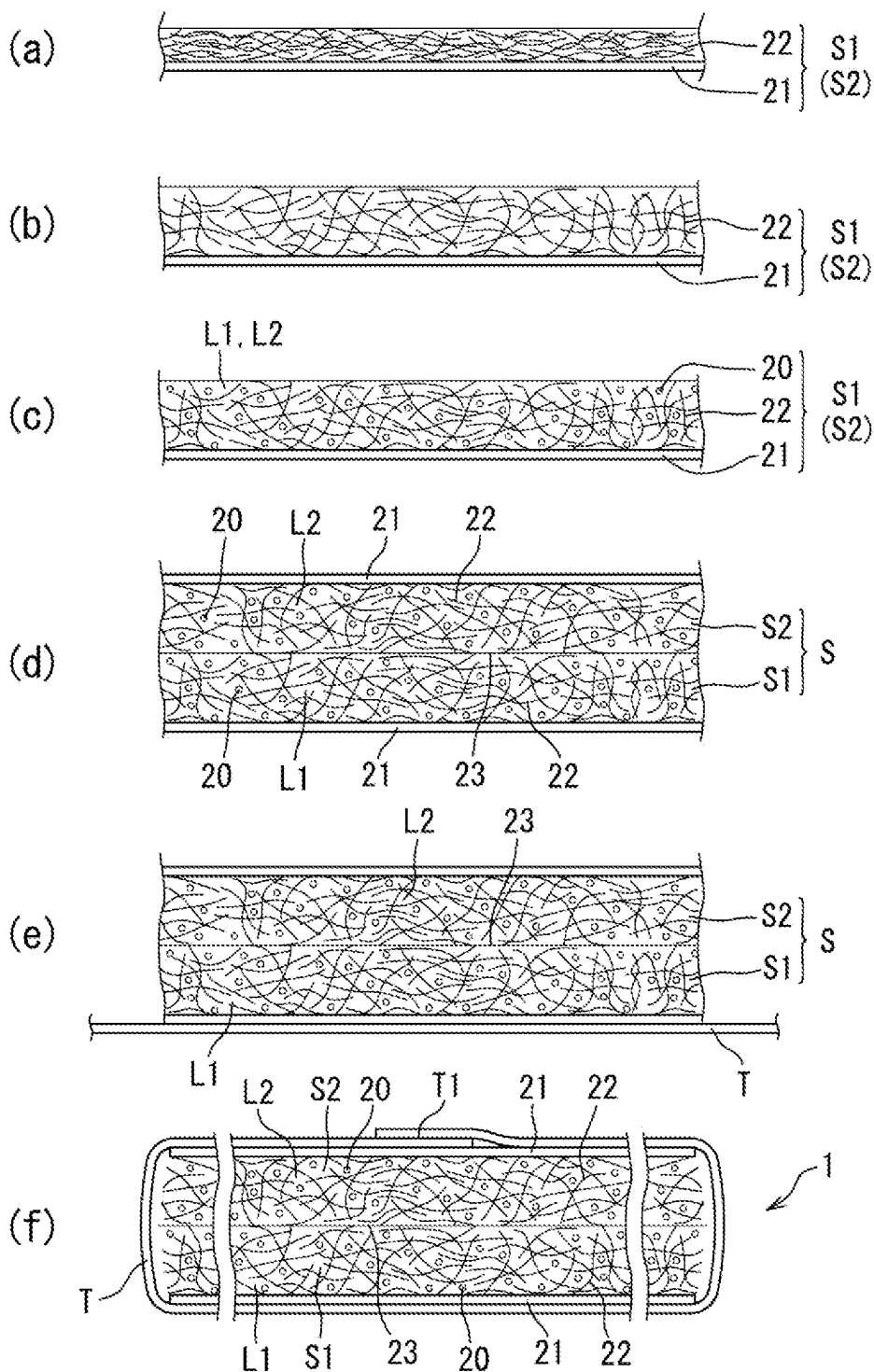


FIG. 4

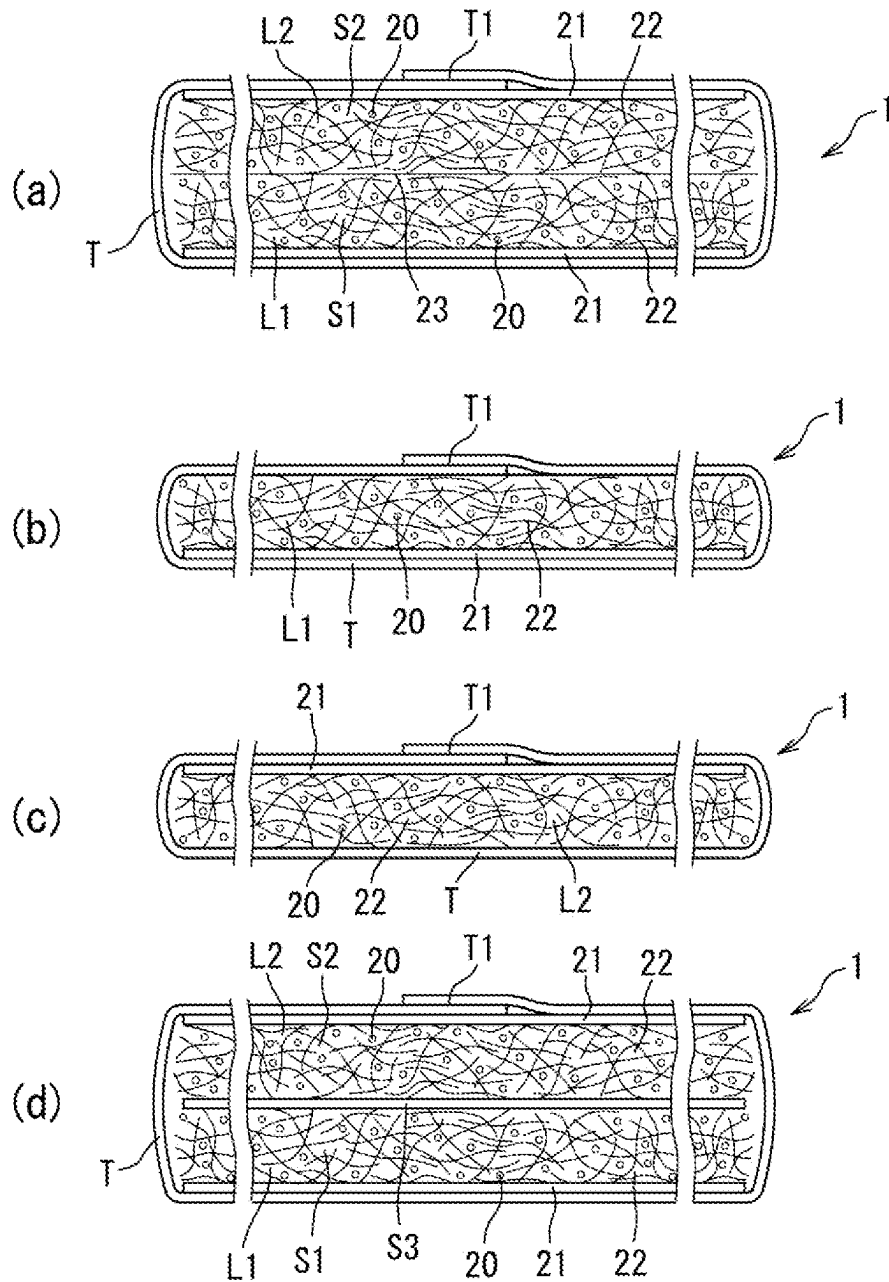


FIG. 5

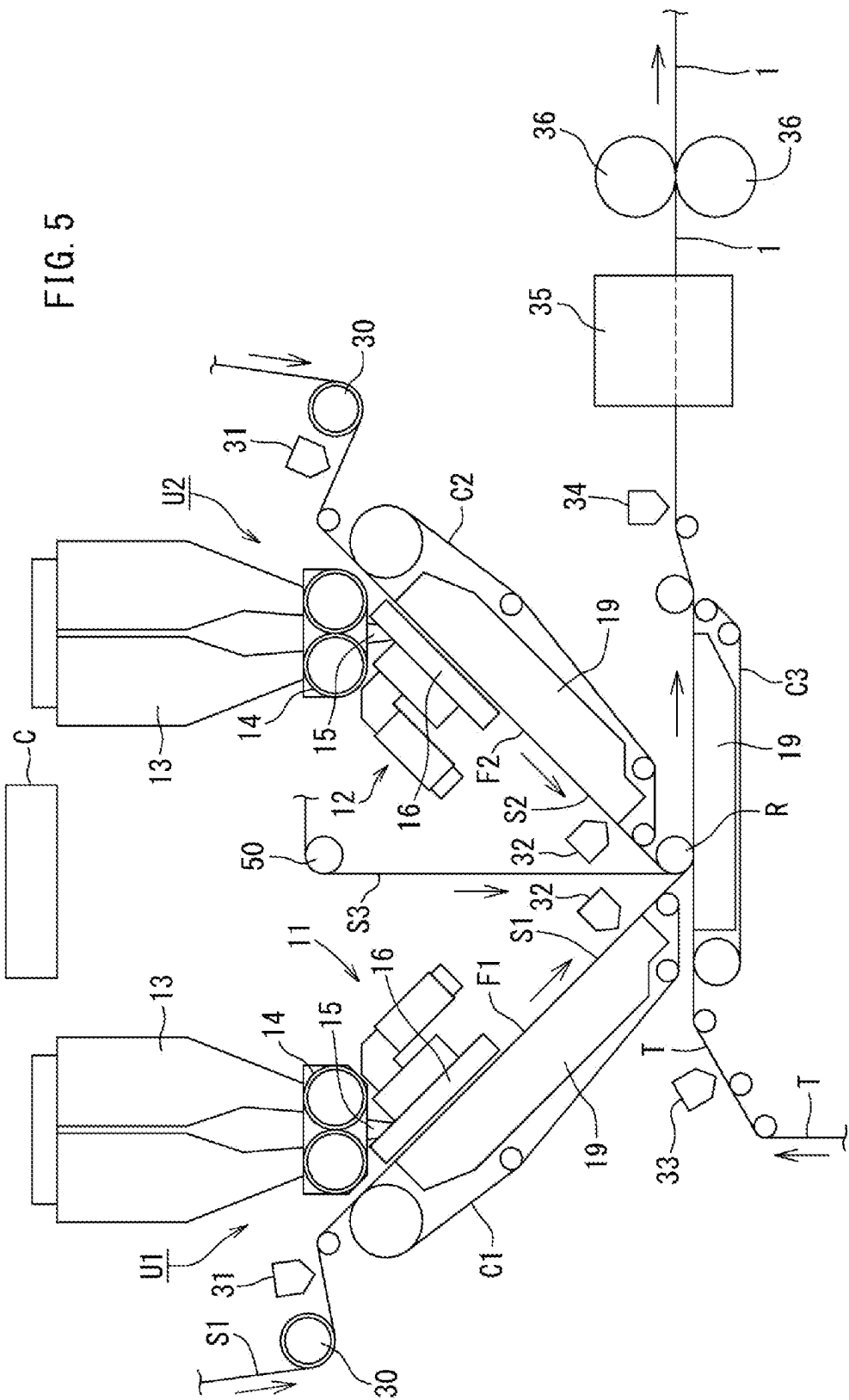
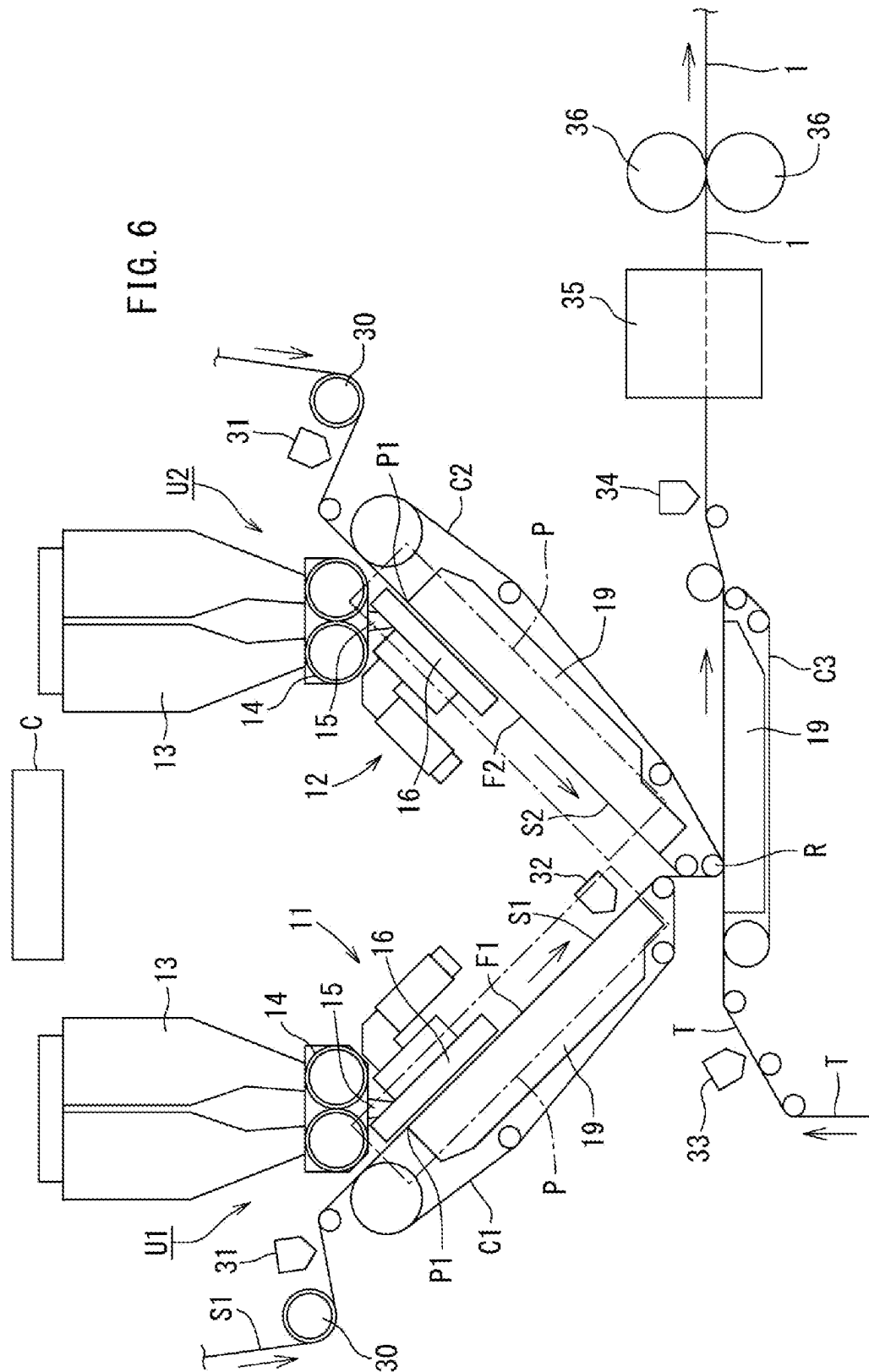


FIG. 6



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MANUFACTURING APPARATUS AND METHOD FOR ABSORBENT BODY

TECHNICAL FIELD

The present invention is related to manufacturing apparatus and method for an absorbent body having super absorbent particle polymer in its fluffed-up sheet.

BACKGROUND ART

Such type of the apparatus and method have been known in the art. (Patent document 1 below) In the invention disclosed in the document 1, the first absorbent layer is formed by sprinkling the absorbent granular powder to one-side surface of the first sheet conveyed by the first drum and keeping the powder in the surface; the second absorbent layer is formed by sprinkling the granular powder to one-side surface of the second sheet conveyed by the second drum and keeping the powder in the surface; and the surfaces of the first and second sheets having the granular powder therein are faced and stacked with each other, thereby the absorbent body formed.

CITATION LIST

Patent Document

Patent document 1: WO2015/123087

SUMMARY OF INVENTION

However, as the granular powder is sprinkled on the cylindrical surface of the drum, centrifugal force generates to each absorbent layer along the cylindrical surface. Thus, the granular powder is likely to scatter around the apparatus.

The object of the present invention is to prevent absorbent granular powder from scattered in manufacturing apparatus and method for an absorbent body.

A manufacturing apparatus of the present invention includes:

- a first conveyor C1 that conveys an air-permeable first sheet S1 while sucking the first sheet S1 along a flat first slope surface F1;
- a second conveyor C2 that conveys an air-permeable second sheet S2 while sucking the second sheet S2 along a flat second slope surface F2 having a slope opposite to the first slope surface F1;
- a first distribution device 11 that forms a first absorbent layer L1 in the first sheet S1 by distributing absorbent powder on an upper surface of the first sheet S1 being conveyed along the first slope surface F1;
- a second distribution device 12 that forms a second absorbent layer L2 in the second sheet S2 by distributing absorbent powder on an upper surface of the second sheet S2 being conveyed along the second slope surface F2; and
- a stacking part that stacks the first sheet S1 conveyed by the first conveyor C1 and the second sheet S2 conveyed by the second conveyor C2 in such a manner as to cause the first absorbent layer L1 of the first sheet S1 and the second absorbent layer L2 of the second sheet S2 to face each other.

Meanwhile, the method of the present invention includes:

- a step of conveying the first and second sheets S1 and S2 toward the stacking part while sucking the first sheet S1 and the second sheet S2 along the flat first slope surface

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of the first conveyor C1 and the flat second slope surface of the second conveyor C2 respectively;

a step of distributing the powder on the upper surfaces of the first and second sheets S1 and S2 extending along the respective flat slope surfaces;

a step of forming the first and second absorbent layers L1 and L2 by supporting the distributed powder on the first and second sheets S1 and S2; and

a step of stacking the first sheet S1 and the second sheet S2 in such a manner as to cause the first absorbent layer L1 and the second absorbent layer L2 to face each other at the stacking part.

In the present invention, the first and second slope surfaces are not curved cylindrical surfaces, but flat surfaces. Thus, the granular powder does not scatter around even centrifugal force is generated. Also, centrifugal force does not generate in the granular powder not sucked by the corresponding conveyor. Thus, such granular powder drops toward the stacking part of both absorbent layers, and is held between the absorbent layers without scattered around.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic layout diagram showing an embodiment 1 of the manufacturing apparatus of the present invention.

FIG. 2 is a side view showing an enlarged stacking part.

FIG. 3 is an enlarged cross-sectional view showing manufacturing process of an absorbent body.

FIG. 4 is an enlarged cross-sectional view showing another structure of an absorbent body.

FIG. 5 is a schematic layout diagram showing an embodiment 2 of the manufacturing apparatus.

FIG. 6 is a layout diagram showing an embodiment 3 of the manufacturing apparatus.

DESCRIPTION OF EMBODIMENTS

In a preferred apparatus, the stacking part is a stacking roller R that rolls in the first sheet S1 conveyed by the first conveyor C1 and the second sheet S2 conveyed by the second conveyor C2 and also stacks the first sheet S1 and the second sheet S2 on each other.

In this case, it is easy to stack the first sheet and the second sheet.

More preferably, the first conveyor C1, the second conveyor C2, and the stacking roller R are arranged in such a manner that the upper surfaces of the first and second sheets S1 and S2 conveyed by the first conveyor C1, the second conveyor C2, and the stacking roller R form a V-shape valley.

In a case where both conveyors and the stacking roller R form a V-shape valley aforementioned, granular powder not held on the surface of each sheet is to be sandwiched between the two absorbent layers.

Another preferred apparatus further includes an introduction part 50 that introduces a liquid-permeable third sheet S3 to be interposed between the first sheet S1 and the second sheet S2 into the stacking roller R.

In this case, the third sheet is provided between the two sheets. Thus, it is possible to prevent gel block—granular powder held in the first and second sheets are joined together—from occurring.

Another preferred apparatus further includes a controller C that selectively controls operation of at least one of a first unit U1 including the first conveyor C1 and the first distri-

bution device **11** and a second unit **U2** including the second conveyor **C2** and the second distribution device **12**.

In this case, it is possible to manufacture absorbent bodies having different thickness and shapes as described later.

Any feature illustrated and/or depicted in conjunction with one of the aforementioned aspects or the following embodiments may be used in the same or similar form in one or more of the other aspects or other embodiments, and/or may be used in combination with, or in place of, any feature of the other aspects or embodiments.

EMBODIMENTS

The present invention will be understood more clearly from the following description of preferred embodiments taken in conjunction with the accompanying drawings. Note however that the embodiments and the drawings are merely illustrative and should not be taken to define the scope of the present invention. The scope of the present invention shall be defined only by the appended claims. In the accompanying drawings, like reference numerals denote like components throughout the plurality of figures.

Prior to an explanation of the present manufacturing apparatus, one example of an absorbent body manufactured by the present manufacturing apparatus is explained first.

FIG. **3** is an enlarged cross-sectional view showing an absorbent body as well as its manufacturing steps. FIG. **3** (f) shows an absorbent body **1** used as an absorbent core for a disposable diaper, for example.

As shown in FIG. **3** (f), an absorbent body **1** includes: a first sheet **S1** and a second sheet **S2** each having granular powder **20**; and a thin paper (tissue paper) **T** wrapping the first sheet **S1** and the second sheet **S2**.

In FIG. **3** (a), the first and second sheets **S1**, **S2** are not yet fluffed up (i.e., before raising). In this FIG., a spread layer **21** at the back side as a base and a short fiber layer **22** at the front side are stacked.

The short fiber layer **22** is a non-woven fabric layer formed from short fiber, and is formed by the air blow process, for example. Such non-woven fabric layer may be formed by blowing hot air on lined-up short fiber.

On the other hand, the spread layer **21** is thinner than the short fiber layer **22** and has higher density than the short fiber layer **22**. The spread layer **21** allows liquid to permeate across an extensive area due to its high diffusibility in the planar direction.

In FIG. **3** (b), the first and second sheets **S1**, **S2** are fluffed up. The first and second sheets **S1**, **S2** transform from a state where the short fiber layer **22** lies (not fluffed-up) to a state where the short fiber layer **22** is raised to have small bulk density and to become thick.

As shown in FIG. **3** (c), the short fiber layer **22** of the first and second sheet **S1**, **S2** after raising is in a state where the granular powder **20** is sprinkled. The granular powder **20** is well-known superabsorbent polymer (SAP). The short fiber layer **22** holding the granular powder **20** configures a first absorbent layer **L1** and a second absorbent layer **L2**.

As shown in FIG. **3** (d), the first and second sheets **S1**, **S2** including the granular powder **20** are stacked together such that the short fiber layers **22** of both sheets contact with each other with an adhesion layer **23** therebetween, thereby becoming one-sheet thick stack **S**. In this state, the first and second absorbent layers **L1**, **L2** face and contact each other.

As shown in FIG. **3** (e)-(f), the stack **S** is wrapped by the thin paper **T**, becoming a well-known absorbent body **1**.

Now, embodiments of the manufacturing apparatus by the present invention are explained with reference to the drawings.

FIG. **1** and FIG. **2** shows an embodiment 1.

As shown in FIG. **1**, the present manufacturing apparatus includes a first unit **U1**, a second unit **U2**, a stacking roller **R** (one example of stacking part), and a controller **C**. The first and second units **U1**, **U2** have a similar structure, and are arranged in reflection symmetry (plane symmetry).

The first unit **U1** includes a first conveyor **C1** and a first distribution device **11**. The second unit **U2** includes a second conveyor **C2** and a second distribution device **12**. The controller **C** selectively controls at least one of the first and second units **U1**, **U2**.

The first conveyor **C1** conveys an air-permeable first sheet **S1** along a flat first slope surface **F1** while sucking the first sheet **S1**. The second conveyor **C2** conveys an air-permeable second sheet **S2** along a flat second slope surface **F2** tilting opposite to the first slope surface **F1** while sucking the second sheet **S2**. As described later, top surfaces (upper surfaces) of the first and second sheets **S1**, **S2** are fluffed up (raised) in advance.

The first distribution device **11** sprinkles absorbent granular powder on the upper surface of the first sheet **S1** conveyed along the first slope surface **F1**, forming the first absorbent layer **L1** in the short fiber layer **22** of the first sheet **S1** of FIG. **2**. The second distribution device **12** of FIG. **1** sprinkles absorbent granular powder on the upper surface of the second sheet **S2** conveyed along the second slope surface **F2**, forming the second absorbent layer **L2** in the short fiber layer **22** of the second sheet **S2** of FIG. **2**. After sprinkled, the granular powder **20** gets into each short fiber layer (absorbent layer) **22** by negative pressure from a corresponding negative pressure case **19** while the first and second sheets **S1**, **S2** are conveyed by the first and second conveyors **C1**, **C2**, respectively.

In FIG. **2**, the stacking roller **R** stacks the first and second sheets **S1**, **S2** so that the first absorbent layer **L1** of the first sheet **S1** conveyed by the first conveyor **C1** and the second absorbent layer **L2** of the second sheet **S2** conveyed by the second conveyor **C2** face each other. In other words, the stacking roller **R** rolls in the first sheet **S1** conveyed by the first conveyor **C1** and the second sheet **S2** conveyed by the second conveyor **C2**, and puts them together.

The first conveyor **C1**, the second conveyor **C2** and the stacking roller **R** are arranged so that the upper surfaces of the first and second sheets **S1**, **S2** conveyed by the first conveyor **C1**, the second conveyor **C2** and the stacking roller **R** of FIG. **1** form a V-shaped valley.

Now, feeding the first and second sheets **S1**, **S2** is described.

As shown in FIG. **1**, the first and second sheets **S1**, **S2** are fed to a corresponding fluff-up roller **30**, and then are conveyed by the respective first and second conveyors **C1**, **C2**. At the stacking roller **R**, the first and second sheets **S1**, **S2** are joined and stacked together. As described later, each conveyor **C1**, **C2** of FIG. **2** is provided with a negative pressure case **19** that sucks a corresponding sheet during conveyance to hold the granular powder **20** in the short fiber layer **22** of the sheet.

The fluff-up roller **30** of FIG. **1** is a roller having a lot of teeth and rotates slower than the conveyance speed of the sheets **S1**, **S2**. With this speed difference, the fluff-up roller **30** raises (fluffs up) the short fiber layer **22** of the sheet of FIG. **3** (a) as shown in FIG. **3** (b).

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The first applicator **31** for adhesive is provided between the fluff-up roller **30** and the conveyors **C1**, **C2** of FIG. 1. Adhesive is applied to the front surface of the raised short fiber layer **22** (FIG. 3 (b)).

Now, sprinkling of the granular powder **20** of FIG. 1 is explained.

A first distribution device **11** and a second distribution device **12** sprinkle the granular powder **20** on the sheet. Each device includes a hopper **13** storing the granular powder **20**, a meter **14**, a guider **15** and a distribution case **16**. The distribution case **16** is provided with a distribution mouth and a shutter for opening/closing the distribution mouth. With this configuration, a predetermined granular powder **20** is sprinkled intermittently on the short fiber layer **22** (FIG. 2) of the corresponding sheet.

Feeding of sheets and sprinkling of the granular powder are disclosed in WO2017/131014, for example, and its disclosure is incorporated herein by reference.

Now, steps of stacking the sheets **S1**, **S2** and thereafter are explained.

As shown in FIG. 2, the second applicator **32** applies adhesive to the sheet **S1** of the sheets, which the granular powder **20** is sprinkled. With this application, the short fiber layers **22** of the first and second sheets **S1**, **S2** are joined together after the sheets **S1**, **S2** are stacked by the stacking roller **R**.

In FIG. 1, a thin paper **T** is fed on the stacking conveyor **C3**. The third applicator **33** is provided upstream of the stacking conveyor **C3**, and applies adhesive to the center area of the upper surface of the thin paper **T**.

As shown in FIG. 2, the stacking conveyor **C3** conveys the sheet material downstream in a state where the thin paper **T** is provided under the stack **S** of the first and second sheets **S1**, **S2** stacked together. In the conveyance, the stack **S** is joined to the front surface of the thin paper **T**. In the present embodiment, the stacking conveyor **C3** contacts the stacking roller **R** with the stack **S** and the thin paper **T** therebetween.

The fourth applicator **34**, a folder **35** and a pair of pressure contact rollers **36** are provided downstream of the stacking conveyor **C3** of FIG. 1. Note that the stacking conveyor **C3** includes a negative pressure case **19** sucking the thin paper **T**.

The fourth applicator **34** applies adhesive to side edges of the upper surface of the thin paper **T**. Note that a width of the thin paper **T** is more than two times larger than that of the stack **S**.

The folder **35** of FIG. 1 folds the thin paper **T** as shown in FIG. 3 (f) to wrap the stack **S** by the thin paper **T**, forming the absorbent body **1**. The pressure contact roller **36** presses hard the absorbent body **1** in its thickness direction.

Now, a method of manufacturing the absorbent body **1** having the first sheet **S1** and the second sheet **S2** of FIG. 3 (f) is explained. In this case where the first and second sheets are used, both first and second units **U1**, **U2** of FIG. 1 are operated.

The first sheet **S1** and the second sheet **S2** are fed to the corresponding fluff-up roller **30,30**, and the short fiber layer **22** of FIG. 3 (a) of each sheet is raised (fluffed up). Then, the first applicator **31** (FIG. 1) applies adhesive to the raised short fiber layer **22** of FIG. 3 (b). Thereafter, the first and second sheets **S1**, **S2** of FIG. 1 are conveyed toward the stacking roller **R** along the flat slope surface of the respective first and second conveyors **C1**, **C2** while the sheets **S1**, **S2** are sucked by the first and second conveyors **C1**, **C2**.

Granular powder is sprinkled on the upper surfaces of the first and second sheets **S1**, **S2** of FIG. 1 along their flat slope surfaces. The first distribution device **11** and the second

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distribution device **12** sprinkle the granular powder **20** on the respective first and second sheet **S1**, **S2**, forming the first absorbent layer **L1** and the second absorbent layer **L2** (FIG. 3 (c)) of FIG. 2 including the granular powder **20**. In other words, the sprinkled granular powder **20** is held in the short fiber layer **22** of the first and second sheets **S1**, **S2**, thereby the first and second absorbent layers **L1**, **L2** formed.

In detail, after sprinkled, the granular powder **20** gets into the short fiber layer (absorbent layer) **22** of each first and second sheet **S1**, **S2** by the negative pressure from the corresponding negative pressure case **19** while the first and second sheets **S1**, **S2** are conveyed by the first and second conveyors **C1**, **C2**.

Thereafter, the first sheet **S1** and the second sheet **S2** are stacked together at the stacking roller **R** so that the first absorbent layer **L1** and the second absorbent layer **L2** face each other. In other words, the first absorbent layer **L1** of the first sheet **S1** and the second absorbent layer **L2** of the second sheet **S2** are stacked in a state where they contact each other.

In the present embodiment, the second applicator **32** applies adhesive to the first absorbent layer **L1**, and then the first absorbent layer **L1** and the second absorbent layer **L2** are pulled into the stacking roller **R**, forming the stack **S** of FIG. 3 (d). The stack **S** is arranged on the thin paper **P** conveyed by the stacking roller **C3** of FIG. 2.

Thereafter, the thin paper **T** is folded by the folder **35** of FIG. 1, and then the absorbent body **1** is pressed hard by the pressure contact roller **36**, thereby the absorbent body **1** of FIG. 3 (f) formed. In this process, as shown in FIG. 4 (a), the first absorbent layer **L1** and the second absorbent layer **L2** contact each other and they are sandwiched between the spread layers **21**, **21**.

The present manufacturing apparatus is able to manufacture the absorbent body **1** of FIG. 4 (b) and FIG. 4 (c) in addition to the absorbent body **1** of FIG. 4 (a).

In a case where the absorbent body **1** of FIG. 4 (b) is manufactured, the first unit **U1** at the left side of FIG. 1 is operated whereas the second unit **U2** at the right side is stopped. In this case, an overlap part **T1** of the thin paper **T** contacts the first absorbent layer **L1**, and the overlap part **T1** and the spread layer **21** are arranged sandwiching the short fiber layer **22** (the first absorbent layer **L1**).

In a case where the absorbent body **1** of FIG. 4 (c) is formed, the second unit **U2** at the right side of FIG. 1 is operated whereas the first unit **U1** at the left side is stopped. In this case, the overlap part **T1** of FIG. 4 (c) of the thin paper **T** contacts the spread layer **21**.

Now, a case where the absorbent body **1** of FIG. 4 (d) is explained.

The absorbent body **1** of FIG. 4 (d) has a liquid-permeable third sheet **S3** between the first sheet **S1** and the second sheet **S2**. The absorbent body **1** is produced by the manufacturing apparatus of FIG. 5.

In FIG. 5, in addition to the manufacturing apparatus of FIG. 1, this embodiment further includes an introduction roller **50** introducing the third sheet **S3** into the stacking roller **R**. The third sheet **S3** is introduced between the first sheet **S1** and the second sheet **S2**.

Preferably, the third sheet **S3** is a sheet having diffusibility as the spread layer **21** of the first and second sheets **S1**, **S2**.

Now, the manufacturing apparatus of the embodiment 3 in FIG. 6 is explained.

The manufacturing apparatus of FIG. 6 is closely similar to that of the embodiment 1 in FIG. 1. Thus, different parts from the embodiment 1 are mainly explained.

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In FIG. 6, negative pressure cases 19 are provided with respective first and second conveyors C1, C2, and may be set to suck air in an area downstream from a point P1 around where the granular powder 20 (FIG. 2) drops. This arrangement prevents the granular powder 20 from scattered by air flow generated between each distribution device 11, 12 and the corresponding conveyor.

Also, scatter-prevention boards P, P may be provided on opposite sides of each conveyor C1, C2 so as to extend along the flow direction of the first and second conveyors C1, C2. The scatter-prevention board P prevents air flow from generated in directions other than the flow direction of the conveyors C1, C2, and thus is useful in preventing scatter of the granular powder 20.

As in this embodiment, the stacking roller R may configure a part of the first conveyor C1 or the second conveyor C2. In this embodiment, the lower-end roller of the second conveyor C2 configures the stacking roller R.

In a case where a diameter of the stacking roller R is large as in FIG. 1, the granular powder 20 (FIG. 2) may scatter near the stacking roller R. With a small diameter of the stacking roller R as in FIG. 6, it is likely to prevent scatter of the granular powder 20.

While a preferred embodiment has been described above with reference to the drawings, various obvious changes and modifications will readily occur to those skilled in the art upon reading the present specification.

For example, if a liquid-permeable non-woven fabric sheet having a fluff-up layer is employed, the fluff-up roller will not need to be provided.

If a pair of nip rollers is employed as the stacking roller R, the first unit U1 and the second unit U2 will be arranged in completely symmetry.

Thus, such changes and modifications shall fall within the scope of the present invention as defined by the appended claims.

INDUSTRIAL APPLICABILITY

The present invention is used in manufacturing various absorbent bodies like disposable underwear and diapers.

REFERENCE SIGNS LIST

1: Absorbent body 11: First distribution device 12: Second distribution device 13: Hopper 14: Meter 15: Guide 16: Distribution case 19: Negative pressure case 20: Granular powder (Powder and granular material) 21: Spread layer 22: Short fiber layer 23: Adhesion layer 30: Fluff-up roller 31-34: Applicator 35: Folder 36: Pressure contact roller 50: Introduction part C1: First conveyor C2: Second conveyor C3: Stacking conveyor F1: First slope surface F2: Second slope surface L1: First absorbent layer L2: Second absorbent layer S: Stack S1: First sheet S2: Second sheet S3: Third sheet T: Thin paper T1: Overlap part U1: First unit U2: Second unit C: Controller P: Scatter-prevention board P1: Point R: Stacking roller

The invention claimed is:

1. A manufacturing apparatus for an absorbent body comprising:

a first conveyor that conveys an air-permeable first sheet while sucking the first sheet along a flat first slope surface;

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a second conveyor that conveys an air-permeable second sheet while sucking the second sheet along a flat second slope surface having a slope opposite to the first slope surface;

a first distribution device that forms a first absorbent layer in the first sheet by distributing absorbent powder on an upper surface of the first sheet being conveyed along the first slope surface;

a second distribution device that forms a second absorbent layer in the second sheet by distributing absorbent powder on an upper surface of the second sheet being conveyed along the second slope surface;

a stacking part that stacks the first sheet conveyed by the first conveyor and the second sheet conveyed by the second conveyor in such a manner as to cause the first absorbent layer of the first sheet and the second absorbent layer of the second sheet to face each other; and a stacking conveyor, on which a paper being fed, that conveys the first and second sheets and the paper in a state where the paper is disposed under a stack of the first and second sheets stacked together,

the stacking part being a stacking roller that rolls in the first sheet conveyed by the first conveyor and the second sheet conveyed by the second conveyor and stacks the first sheet and the second sheet on each other, and

the stacking conveyor contacting the stacking roller with the stack and the paper therebetween.

2. The manufacturing apparatus for an absorbent body according to claim 1, wherein

the first conveyor, the second conveyor, and the stacking roller are arranged in such a manner that the upper surfaces of the first and second sheets conveyed by the first conveyor, the second conveyor, and the stacking roller form a V-shape valley.

3. The manufacturing apparatus for an absorbent body according to claim 1, further comprising:

an introduction part that introduces a liquid-permeable third sheet to be interposed between the first sheet and the second sheet into the stacking roller.

4. The manufacturing apparatus for an absorbent body according to claim 1, further comprising:

a controller that selectively controls operation of at least one of a first unit including the first conveyor and the first distribution device and a second unit including the second conveyor and the second distribution device.

5. A manufacturing method for an absorbent body using the manufacturing apparatus for an absorbent body according to claim 1, comprising:

a step of conveying the first sheet toward the stacking roller while sucking the first sheet along the flat first slope surface of the first conveyor;

a step of conveying the second sheet toward the stacking roller while sucking the second sheet along the flat second slope surface of the second conveyor;

a step of distributing the absorbent powder on the upper surfaces of the first sheet extending along the first slope surface;

a step of distributing the absorbent powder on the upper surfaces of the second sheet extending along the second slope surface;

a step of forming the first absorbent layer by supporting the distributed powder on the first sheet;

a step of forming the second absorbent layer by supporting the distributed powder on the second sheet;

a step of stacking the first sheet and the second sheet in such a manner as to cause the first absorbent layer and the second absorbent layer to face each other at the stacking roller; and

a step of feeding the paper to the stacking conveyor and conveying the first and second sheets and the paper in a state where the paper is disposed under the stack of the first and second sheets stacked together and the stacking conveyor contacts the stacking roller with the stack and the paper therebetween.

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